17.1.1 Describing Oscillations

Your notes

Describing Oscillations

- An **oscillation** is defined as:

  Repeated back and forth movements on either side of any equilibrium position

- When the object stops oscillating, it returns to its equilibrium position
  - An **oscillation** is a more specific term for a vibration
  - An **oscillator** is a device that works on the principles of oscillations
- Oscillating systems can be represented by displacement-time graphs similar to transverse waves
- The shape of the graph is a sine curve
  - The motion is described as **sinusoidal**

Properties of Oscillations

- **Displacement (x)** of an oscillating system is defined as:

  The distance of an oscillator from its equilibrium position

- **Amplitude (x_0)** is defined as:

  The maximum displacement of an oscillator from its equilibrium position

- **Angular frequency (ω)** is defined as:

  The rate of change of angular displacement with respect to time

- This is a scalar quantity measured in **rad s^{-1}** and is defined by the equation:

$$\omega = \frac{2\pi}{T} = 2\pi f$$

- **Frequency (f)** is defined as:

  The number of complete oscillations per unit time

- It is measured in Hertz (Hz) and is defined by the equation:

$$f = \frac{1}{T}$$

- **Time period (T)** is defined as:

  The time taken for one complete oscillation, in seconds

- One complete oscillation is defined as:

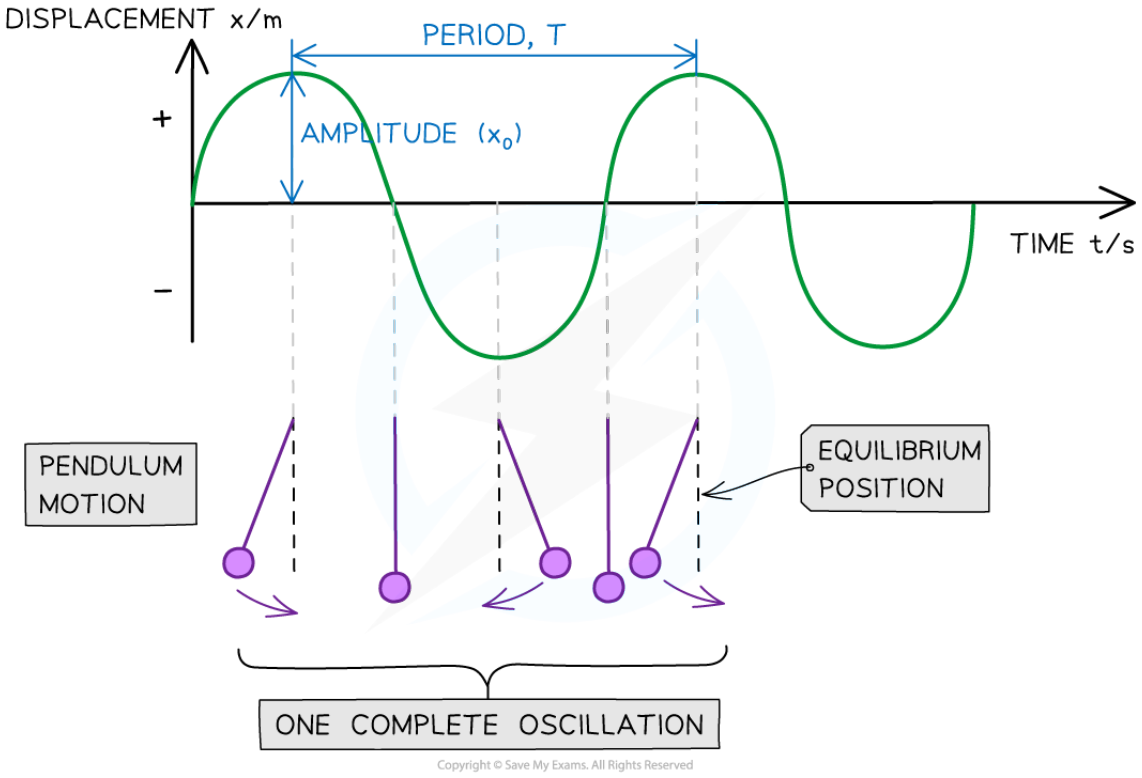
The time taken for the oscillator to pass the equilibrium from one side and back again fully from the other side

Your notes

- The time period is defined by the equation:

$$T = \frac{1}{f} = \frac{2\pi}{\omega}$$

- **Phase difference** is how much one oscillator is in front or behind another
  - When the relative position of two oscillators are equal, they are **in phase**
  - When one oscillator is exactly half a cycle behind another, they are said to be in **anti-phase**
  - Phase difference is normally measured in radians or fractions of a cycle
  - When two oscillators are in antiphase they have a phase difference of **π radians**

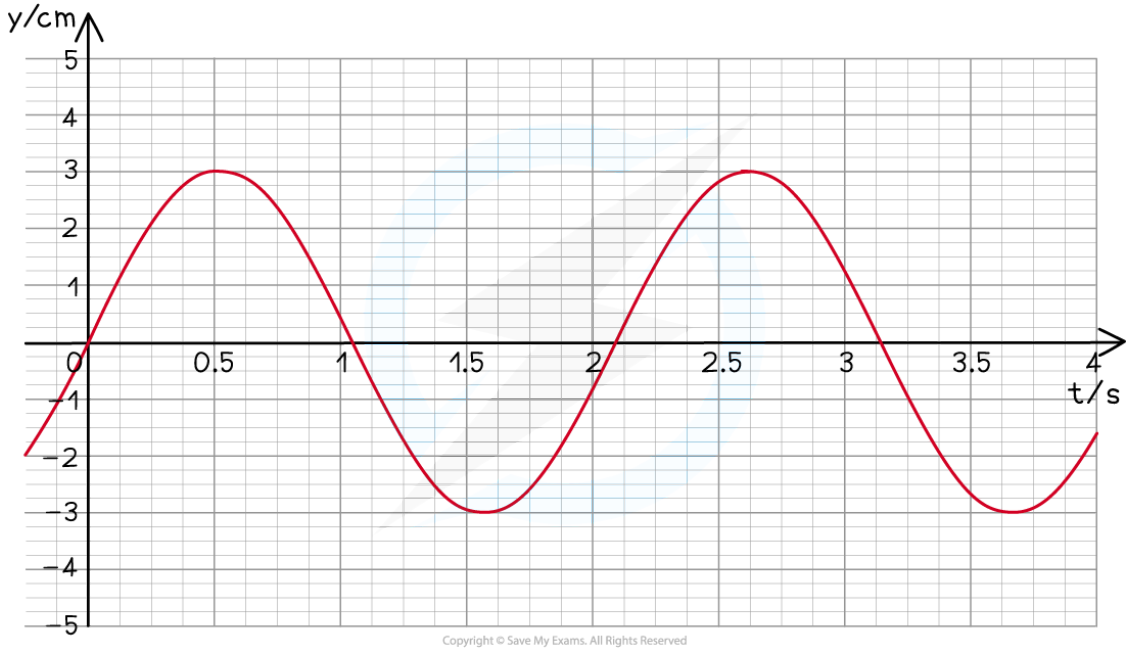


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Displacement-time graph of an oscillation of a simple pendulum

Worked example

A student sets out to investigate the oscillation of a mass suspended from the free end of a spring. The mass is pulled downwards and then released. The variation with time *t* of the displacement *y* of the mass is shown in the figure below.



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Use the information from the figure to calculate the angular frequency of the oscillations.

Step 1:

Write down the equation for angular frequency

$$\omega = \frac{2\pi}{T}$$

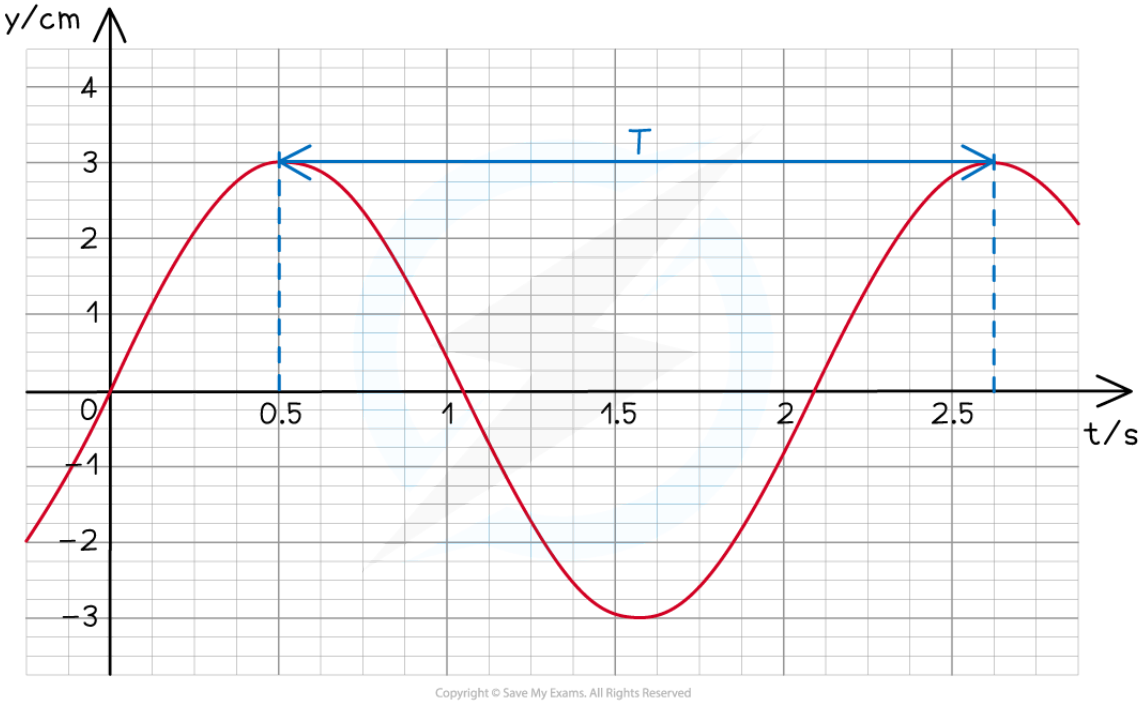
Step 2:

Calculate the time period T from the graph

The time period is defined as the time taken for **one complete oscillation**

This can be read from the graph:

Your notes



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$T = 2.6 - 0.5 = 2.1\ s$

Step 3:

Substitute into angular frequency equation

$$\omega = \frac{2\pi}{2.1} = 2.9919... = 3.0\ rad\ s^{-1}$$

Examiner Tip

The properties used to describe oscillations are very similar to transverse waves. The key difference is that oscillators do not have a 'wavelength' and their direction of travel is only kept within the oscillations themselves rather than travelling a distance in space.

Your notes

17.1.2 Simple Harmonic Motion


Your notes

Conditions for Simple Harmonic Motion

- **Simple harmonic motion (SHM)** is a specific type of oscillation
- SHM is defined as:

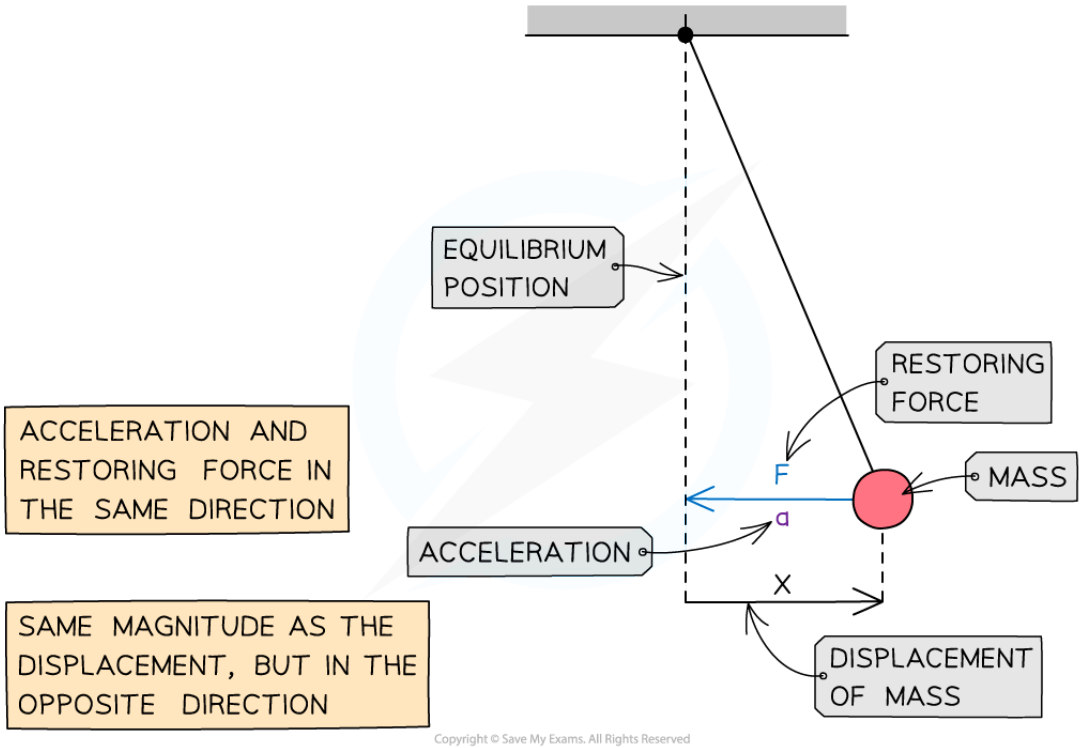
> ***A type of oscillation in which the acceleration of a body is proportional to its displacement, but acts in the opposite direction***

- Examples of oscillators that undergo SHM are:
  - The pendulum of a clock
  - A mass on a spring
  - Guitar strings
  - The electrons in alternating current flowing through a wire
- This means for an object to oscillate specifically in SHM, it must satisfy the following conditions:
  - Periodic oscillations
  - Acceleration proportional to its displacement
  - Acceleration in the opposite direction to its displacement
- Acceleration *a* and displacement *x* can be represented by the defining equation of SHM:

$$a \propto -x$$

- An object in SHM will also have a restoring force to return it to its equilibrium position
- This restoring force will be directly proportional, but in the **opposite direction**, to the displacement of the object from the equilibrium position
- **Note:** the restoring force and acceleration act in the **same direction**

Your notes



Force, acceleration and displacement of a pendulum in SHM

- This is why a person jumping on a trampoline is not an example of simple harmonic motion:
  - The restoring force on the person is **not** proportional to their distance from the equilibrium position
  - When the person is not in contact with the trampoline, the restoring force is equal to their weight, which is constant
  - This does not change, even if they jump higher

17.1.3 Calculating Acceleration & Displacement in SHM

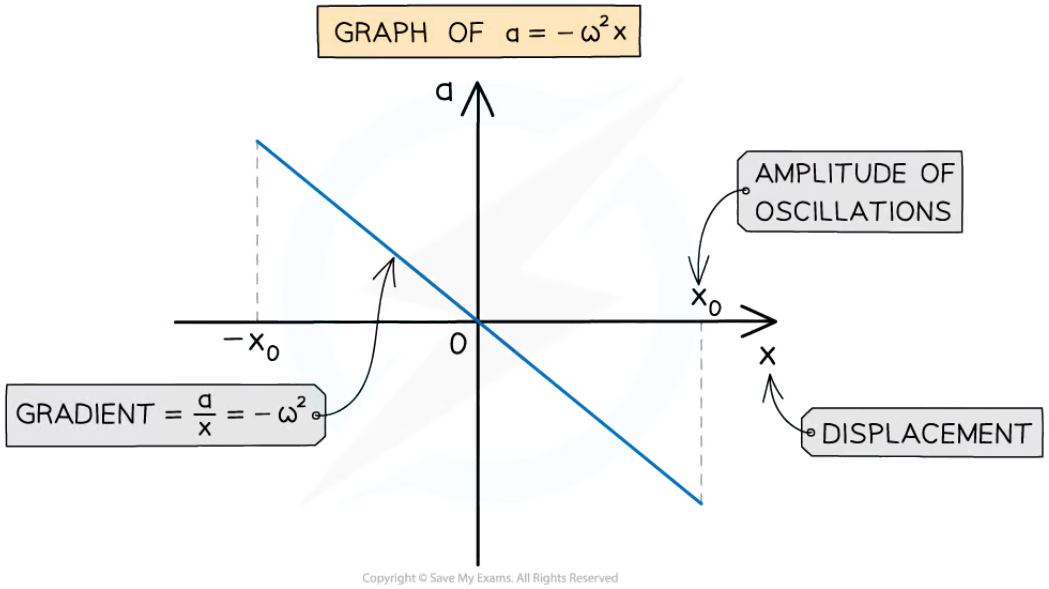
Your notes

Acceleration & Displacement of an Oscillator

- The acceleration of an object oscillating in simple harmonic motion is:

$$a = -\omega^2 x$$

- Where:
  - a = acceleration ($m\ s^{-2}$)
  - ω = angular frequency ($rad\ s^{-1}$)
  - x = displacement (m)
- This is used to find the acceleration of an object in SHM with a particular angular frequency ω at a specific displacement x
- The equation demonstrates:
  - The acceleration reaches its maximum value when the displacement is at a maximum ie. $x = x_0$ (amplitude)
  - The minus sign shows that when the object is displacement to the **right**, the direction of the acceleration is to the **left**



The acceleration of an object in SHM is directly proportional to the negative displacement

- The graph of acceleration against displacement is a straight line through the origin sloping downwards (similar to y = – x)
- Key features of the graph:
  - The gradient is equal to $-\omega^2$

Your notes

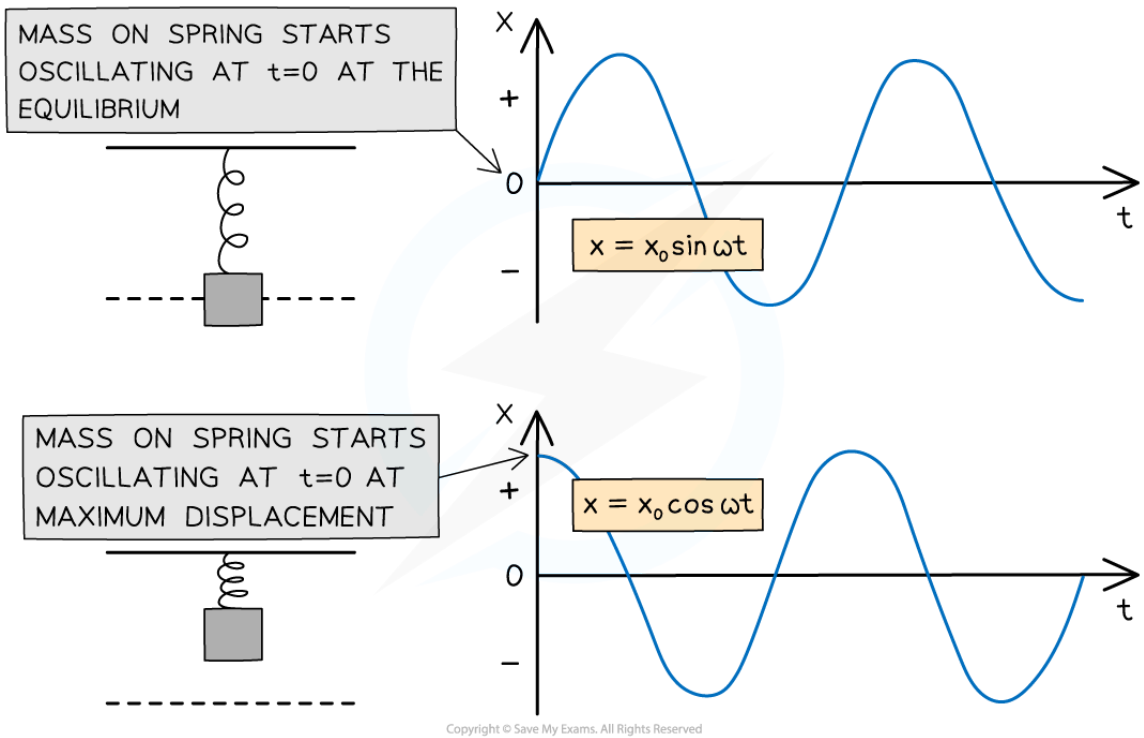
- The maximum and minimum displacement x values are the amplitudes $-x_0$ and $+x_0$
- A solution to the SHM acceleration equation is the displacement equation:

$$x = x_0 \sin(\omega t)$$

- Where:
  - x = displacement (m)
  - x_0 = amplitude (m)
  - t = time (s)
- This equation can be used to find the position of an object in SHM with a particular angular frequency and amplitude at a moment in time
  - **Note:** This version of the equation is only relevant when an object begins oscillating from the equilibrium position (x = 0 at t = 0)
- The displacement will be at its maximum when $\sin(\omega t)$ equals 1 or – 1, when $x = x_0$
- If an object is oscillating from its amplitude position ($x = x_0$ or $x = -x_0$ at t = 0) then the displacement equation will be:

$$x = x_0 \cos(\omega t)$$

- This is because the cosine graph starts at a maximum, whilst the sine graph starts at 0



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These two graphs represent the same SHM. The difference is the starting position

Your notes

Worked example

A mass of 55 g is suspended from a fixed point by means of a spring. The stationary mass is pulled vertically downwards through a distance of 4.3 cm and then released at t = 0.

The mass is observed to perform simple harmonic motion with a period of 0.8 s.

Calculate the displacement x in cm of the mass at time t = 0.3 s.

Step 1: Write down the SHM displacement equation

- Since the mass is released at t = 0 at its maximum displacement, the displacement equation will be with the cosine function:

$$x = x_0 \cos(\omega t)$$

Step 2: Calculate angular frequency

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.8} = 7.85 \text{ rad s}^{-1}$$

- Remember to use the value of the time period given, not the time where you are calculating the displacement from

Step 3: Substitute values into the displacement equation

$$x = 4.3 \cos(7.85 \times 0.3) = -3.0369\ldots = \mathbf{-3.0\ cm}\ (2\text{ s.f})$$

- Make sure the calculator is in **radians mode**
- The negative value means the mass is 3.0 cm on the opposite side of the equilibrium position to where it started (3.0 cm above it)

Examiner Tip

Since displacement is a vector quantity, remember to keep the minus sign in your solutions if they are negative, you could lose a mark if not!

Also, remember that your calculator must be in **radians** mode when using the cosine and sine functions. This is because the angular frequency ω is calculated in rad s^{-1}, **not** degrees.

You often have to convert between time period *T*, frequency *f* and angular frequency ω for many exam questions – so make sure you revise the equations relating to these.